IruBasha

A bilingual Vocabulary Learning Web Platform for Sri Lankan Nursery Students

Diploma in Computer Science with Artificial Intelligence

Final Project-Report

DCSAI-251-F



School of Computing and Engineering

National Institute of Business Management

Sri Lanka

Diploma in Computer Science with Artificial Intelligence

FINAL PROJECT REPORT

Project Title: IruBasha

Group : 06

Submitted by:

Fathima Imam – KIC-DCSAI-251-F-001

Dilmini Dasunika – KIC-DCSAI-251-F-003

Sewmini Dissanayake– KIC-DCSAI-251-F-014

Piratheesha Ambikaipakan – KIC-DCSAI-251-F-040

Submitted to:

National Institute of Business Management

Date : 12th January 2026

Declaration

We certify that this project report titled “IruBasha: does not incorporate,without acknowledgement,any material previously submitted for a Diploma or Degree in any institution and,to the best of our knowledge and belief,it does not contain any material previously published or written by another person except where due reference is made in the text.

We also hereby give consent for our project report,if accepted,to be made available for photocopying and for inter-library loans,and for the title and summary to be made available to outside orgarnizations.

Name	Signature	Date
Fathima Imam	_____	_____
Dilmini Dasunika	_____	_____
Sewmini Dissanayake	_____	_____
Piratheesha Ampikaipakan	_____	_____

Acknowledgement

We would like to express our sincere gratitude for our project supervisor Mrs.Sathya Wijewardhana for the continuous guidance,encouragement,and valuable feedback provided throughout the development of this project

We also thank the lecturers of the School of Computing and Engineering at the National Institute of the Business Management for the knowledge and support provided during our academic journey.

Finally ,we would like to thank our families and friends for their encouragement and support throughout the successful completion of this project

Preamble

Abstract

Irubasha is a web-based lingual learning platform developed to support Sinhala and Tamil language learning for Sri Lankan children aged between 3-12 years.In the Sri lankan education context,early language learners often struggle with pronunciation,vocabulary retention and structuring sentences due to lack of interactive digital tools.Existing applications focuses mainly on direct translation and do not provide an nteractive learning space.

To address this gap,IruBasha integrates logic-based AI using SWI-Prolog with modern web technologies.The system follows a client-server architecture where a browser-based frontend communicates with a Node.js backend through RESTful APIs.A Prolog-based dictionary provides explainable,rule -driven translations,while a neural translational service is used as a fallback.The application allows learners to also liste,record and compare their pronunciation.

The platform also includes quizzes,a personal word bank,and basic progress tracking to encourage continuous learning.This project demonstrates how logic programming and hybrid AI approaches can be effectively applied in educational technology for low-resource languages

Keywords

Bilingual Learning,Aritfiicial Intelligence,Prolog,Language Translation,Educational Technology,Sinhala,Tamil,Web-Based Learning

List of Figures

Figure 1: System Architecture Diagram

Figure 2: UML Diagram

Figure 3: Use case Diagram

Figure 4 : ER Diagram

Figure 5: Welcome and Login Interface

Figure 6: Translation Interface

Figure 7: Quiz Interface

Figure 8: Grammer and Daily Challenge Interface

LIST OF TABLES

Table 4.5 : Database Table

Table 4.8 : Report Layout Summary Table

Acronym	Description
AI	Artificial Intelligence
UI	User Interface
API	Application Programming Interface
SQL	Structured Query Language
TTS	Text-to-Speech
PWA	Progressive Web Application

CHAPTER 1-INTRODUCTION

1.1 Introduction of the Software Product

IruBasha is a web-based bilingual learning application designed to assist Srilankan children in learning Sinhala and Tamil in an interactive and engaging manner.The system focuses on vocabulary building,pronunciation practice,and basic grammar awareness through a child-friendly interface.

The application is accessible through a web browser,eliminating the need for software installation.This decision was made to ensure ease of access across school computers,personal devices,especially in environment with low technical resources.

1.2 System Usage Overview

The system consists of two main components: a frontend and a backend.The frontend runs in the user's web browser and is responsible for presenting content,capturing user input and displaying feedback.The backend runs on a Node.js server and handles tranlslation logic,Prolog execution,database operations,and other services.

1.3 Existing Systems and Market Practices

Currentplatforms in Srilanka use printed materials,and generaldigital tools such as only translation and video platforms.While these tools provide a learning capability, they do not allow children to learn,practice and adapt through pronunciation.

Generic translation tools do not explain sentence structure,provide guided practice and support self learning.These limitations motivated the development of a customer and interactive educational platform that also focuses on explainablity,engagement and local language support.

1.4 Target Users

The primary users are children from age 3-12 who are part of early education.Secondary users include parents and teachers who may guide or monitor the learning process.

1.5 Problem Definition

There is a lack of engaging,child-friendly bilingual learning platforms that support Sinhala and Tamil with structured vocabulary learning and pronunciation practice.Existing solutions are too direct,lack of local relevance and less interactive.

1.6 Objectives of the Project

- To develop a bilingual learning platform for Sinhala and Tamil
- To integrate logic-based AI for explainable translations
- To provide pronunciation practice through audio playback and recording.
- To design a child-centric and engaging user interface

1.7 Proposed Solution

The proposed solution is a browser-based bilingual learning platfrom that integrates SWI-Prolog for rule-based dictionary management and Node.js for backend processing.

CHAPTER 2-METHODOLOGY

2.1 Introduction

This chapter talks about methodology in designing and developing the system.It explains the development approach,data collection methods,tools used,and the reasoning behind key technical decisions.

2.2 Data Collection Methods.

Vocabualary data and linguistic structures were collected from AI.

2.3 Software Development Model

The Iterative Waterfall Model is chosen for this project because it provides a structured and well defined development process while still allowing feedback and improvements between stages. This project involves several sequential phases requirement analysis, design, implementation, testing, and deployment, each of which benefits from clear documentation and milestone tracking. However, since feedback from teachers, parents, and students may lead to refinements, the iterative approach allows limited revisiting of previous phases when necessary.

2.4 Development Tools and Technologies

Node.js was selected as the backend platfrom due to its asynchronous,non-blocking architecture, which is well suited to handle multiplenn APIs alternatively.SWI_Prolog was chosen to implement logic-based translation and quiz generation.

SQL-Lite was used as the database management system because it is lightweight,easy to deploy and sufficient for academic-scale applications.HTML+CSS and JacaScript were used for frontend development.

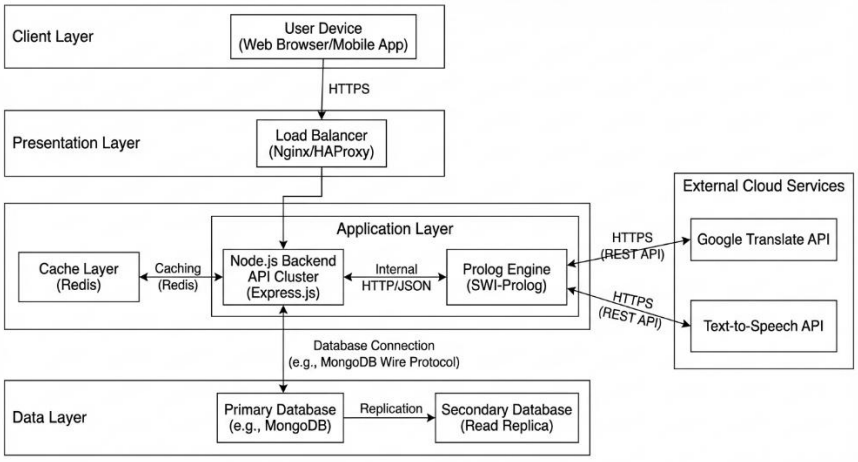
2.5 Testing Methodology

Testing was conducted using a combination of manual testing and Functional testing.

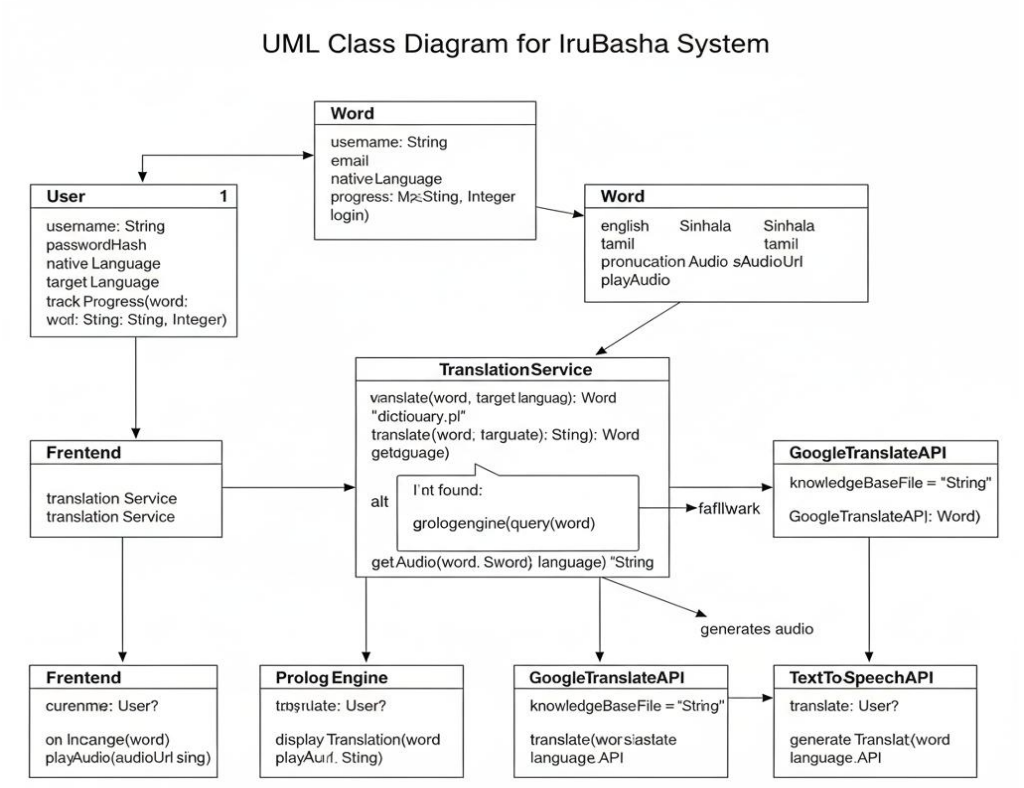
2.6 Implementation Plan

The system was implemented in phased,beginning with basic translation functionality,followed by pronunciation,quizzes,daily challengesmauthentication and data persistence.

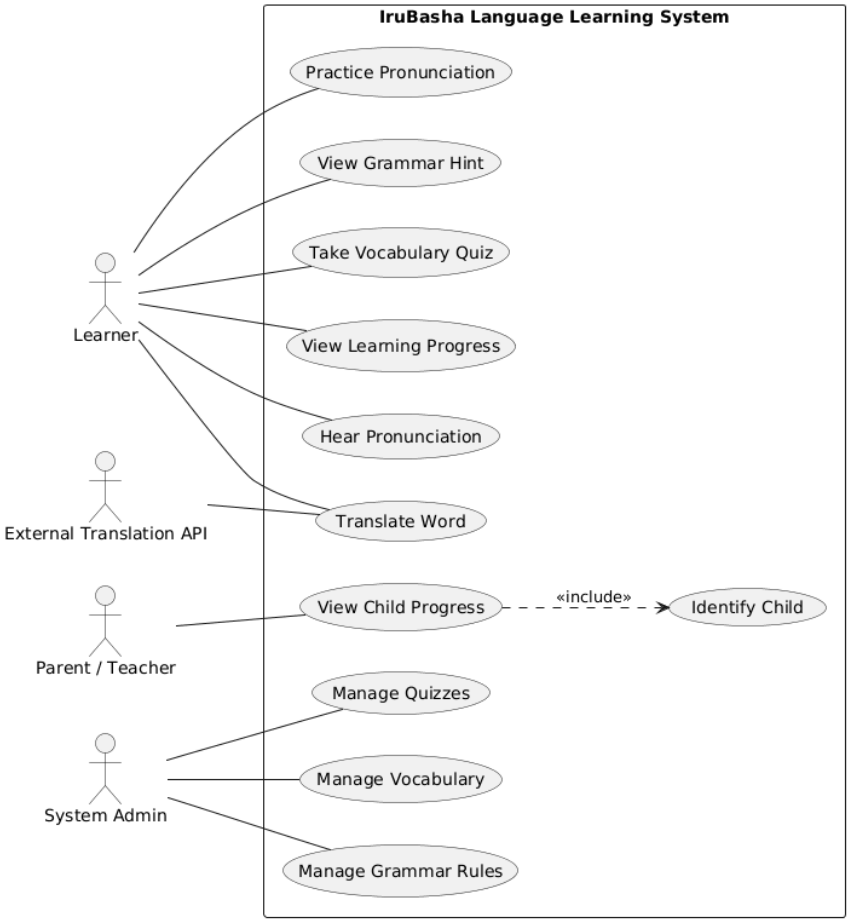
CHAPTER 3- SYSTEM ANALYSIS AND DESIGN



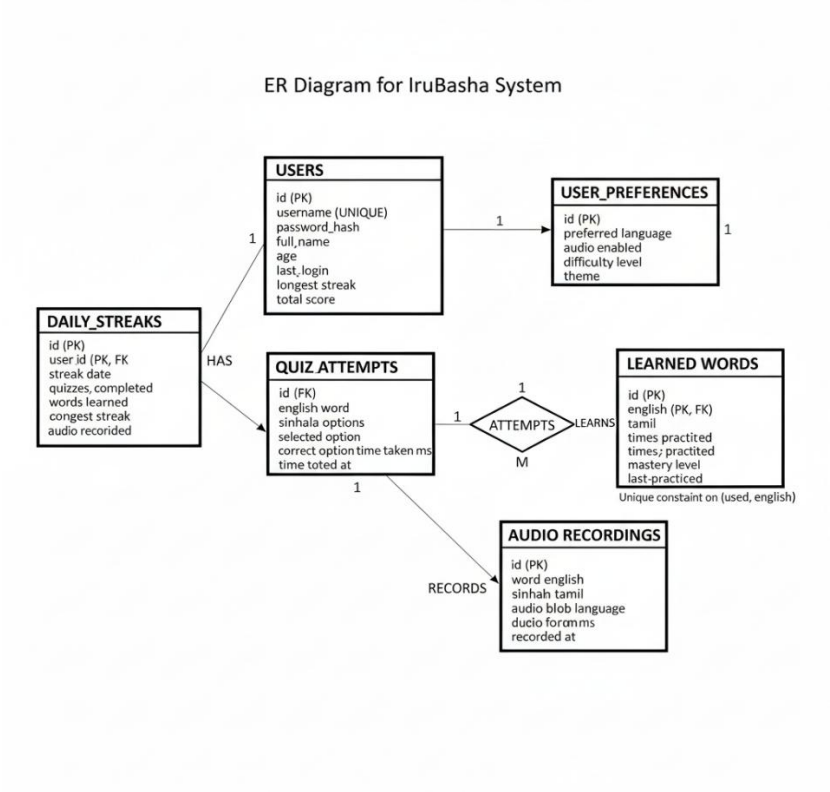
The system follows a client-server architecture where frontend communicates with the backend through RESTful APIs



The class diagram represents entities such as User,Word,and Statistics.



The functional requirements include user authentication,word translation,pronunciation practice,quiz generation,word bank management,and progress tracking.



The database design ensures structured storage of user and learning data.

CHAPTER 4 – SOLUTION DESIGN AND IMPLEMENTATION

4.1 Introduction

This chapter explains how the proposed design was implemented in practice and describes the structure of the frontend,backend,AI logic,database,and communication mechanisms.

4.2 Frontend Design

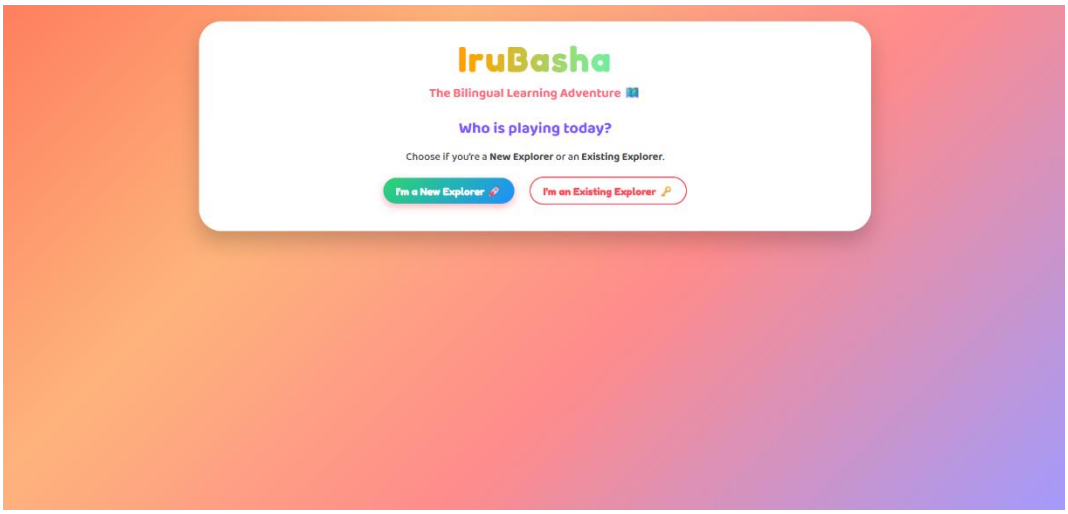
The frontend was implemented using HTML,CSS,and JavaScript.A browser-based interface was chosen to ensure zero installation requirements and maximum accessibility.The interface uses large buttons,readable fonts,and visual feedback to maintain engagement while reducing cognitive load for children,

CSS Flexbox and Grid were used for layout design due to their responsiveness and simplicity.

Interface 4.2.1

Interface:Welcome/Login Screen

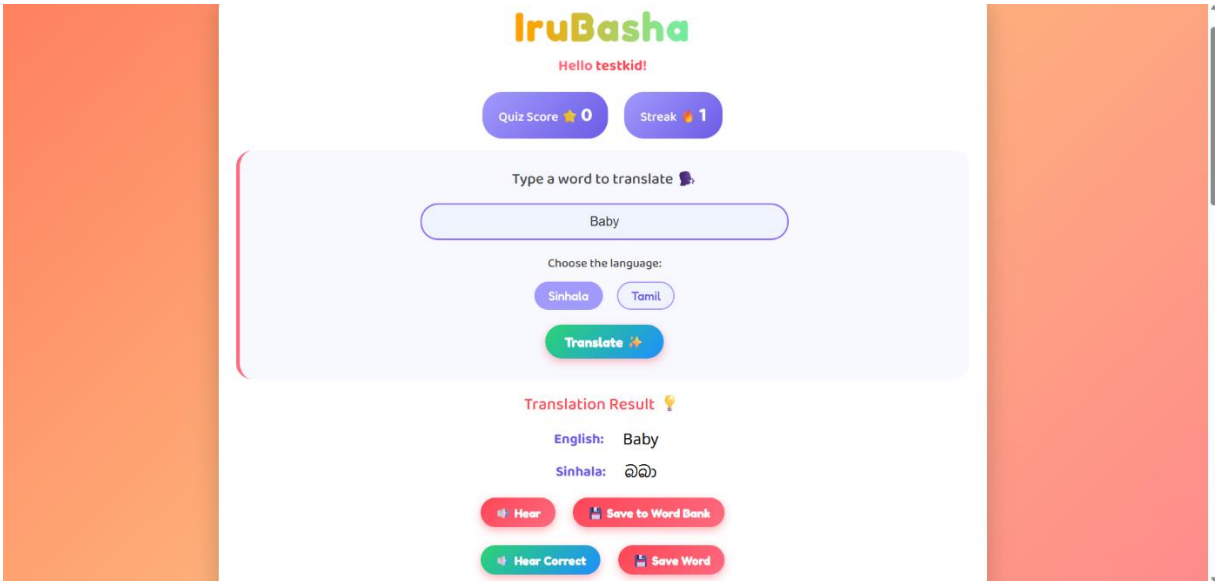
This is the initial screen of IruBasha.It allows users to choose between creating a new explorer profile or logging in as an existing user.



Interface 4.2.2

Interface Name:Translation Interface

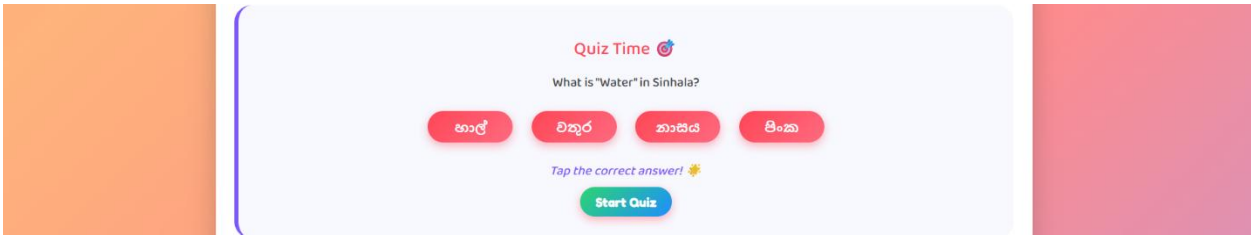
This interface allows a user to enter an English word and receive its Sinhala or Tamil translation.Users can listen to the correct pronunciation,save the word into word bank and pronunciate through audio recording.



Interface 4.2.3

Interface Name:Quiz Interface

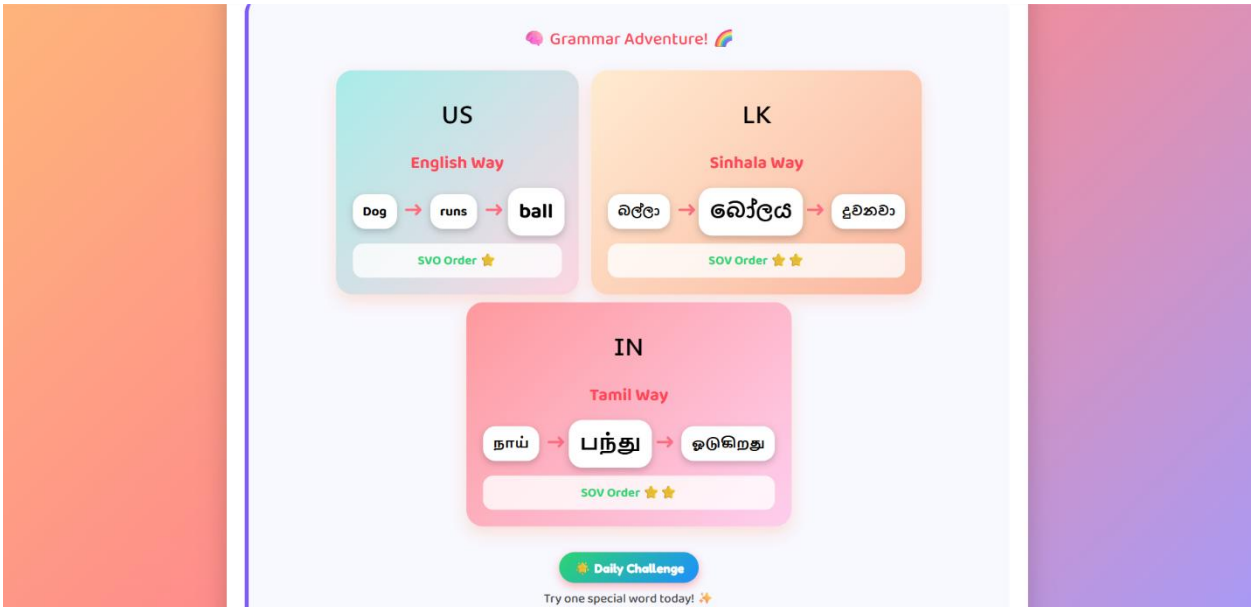
This interface provides multiple-choice quiz questions generated by the system.



Interface 4.2.4

Interface Name : Grammer and Daily Challenge interface

This interace has a grammar learning part and a Daily challenge part where you get to learn new words everyday



4.3 Backend Design

The backend was developed using Node.js with Express framework.Node.js was chosen because it efficiently handles concurrent requests such as translation queries,quiz generation,and database operations.

The backend acts as the central coordinator,managing communication between the frontend,Prolog engine,database and external services.

4.4 Prolog-Based AI Design

SWI-Prolog was used to implement the logic-based dictionary and quiz generation rules.Dictionary entries are defined as Prolog facts,and rules are used to retrieve translations and generate quiz options.

4.5 Database Design

SQLite is used to store userprofiles,saved words,quiz attempts,and learning statistics.

Table No	Table Name	Primary Key	Foreign Key(s)	Description
4.3.1	Users	id	—	Stores registered user information including username, email, age, grade, authentication data, quiz scores, and learning streaks. Acts as the central table for all user-related activities.

4.3.2	Quiz_Attempts	id	user_id → Users(id)	Records each quiz attempt made by a user, including the English word, answer options, selected answer, correctness, and time taken. Used to evaluate learning performance.
4.3.3	Learned_Words	id	user_id → Users(id)	Stores vocabulary words saved by users along with Sinhala and Tamil translations. Tracks practice frequency, mastery level, and last practiced date for personalized learning.
4.3.4	Audio_Recordings	id	user_id → Users(id)	Stores audio recordings created during pronunciation practice. Includes word, language, audio format, duration, and timestamp for playback and feedback.
4.3.5	Daily_Streaks	id	user_id → Users(id)	Maintains daily learning activity records such as quizzes completed, words learned, and audio recordings. Used to calculate learning streaks and encourage consistency.
4.3.6	User_Preferences	user_id	user_id → Users(id)	Stores user-specific settings including preferred language, audio preferences, difficulty level, and theme to personalize the learning experience.

4.6 Authentication and Security

Authentication is implemented using JSON Web Tokens(JWT).Tokens are issued after successful logic and included in subsequent API requests.

4.7 Frontend-Backend Communication

Communication between frontend and backend is implemented using restful APIs over HTTP.The frontend uses the FETCH API to send requests,and the backend responds with JSON data.

4.8 Report Layout Design

The system generates dynamic report layouts within the web interface to present user progress and learning feedback.These reports are designed to be simple,child-friendly,and updated in real time.

Report Layout No	Report Layout Name	Description
4.4.1	Quiz Score and Streak Report	Displays the learner's total quiz score, current streak, and longest streak. This report is updated automatically after each quiz attempt to motivate learners through visible progress indicators.
4.4.2	Word Bank Report	Presents a list of vocabulary words saved by the user, including English, Sinhala, and Tamil translations. Allows learners to review previously learned words and reinforce vocabulary retention.
4.4.3	Pronunciation Practice Feedback Report	Provides feedback after pronunciation practice, including confirmation messages, visual indicators, and audio playback options. Helps learners self-evaluate their pronunciation.
4.4.4	Daily Learning Activity Report	Summarizes daily learning activities such as quizzes completed, words learned, and audio recordings made. Supports learning streak calculation and encourages consistent engagement.

CHAPTER 5- CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project successfully developed IruBasha.The system shows how explainable AI techniques can be combined with modern web technologies to create an educationally relevant application.

5.2 Limitations

The system includes a limited vocabulary under Prolog and does not perform automated pronunciation scoring presently.Browser speech may vary across other platforms.

5.3 Future Enhancements

Future enhancements include explanding the dictionary,implemented automated pronunciation assessment,adding teachers's dashboards and improving adaptive learning features.

Appendices

Appendix A : Project Schedule



Appendix B : Meeting minutes

Date	Attendees	Agenda	Decisions Taken
03/11/2025	All team members	Project introduction and requirement analysis	Identified core problem of bilingual learning and defined system objectives
10/11/2025	All team members	Requirement analysis discussion	Finalized functional requirements and target user age group
17/11/2025	All team members	UI design planning	Decided on child-friendly colors, large buttons, and simple navigation
24/11/2025	All team members	UI design and rule drafting	Finalized interface layouts and basic grammar rule structure
01/12/2025	All team members	Backend architecture discussion	Decided to use Node.js with Prolog integration for translation logic
08/12/2025	All team members	Backend and Prolog integration	Implemented translation flow and fallback logic
15/12/2025	All team members	Trilingual and audio feature discussion	Finalized pronunciation playback and recording features
22/12/2025	All team members	Testing and documentation planning	Identified bugs and started preparing report documentation
29/12/2025	All team members	System testing and improvements	Completed functional testing and minor UI refinements
05/01/2026	All team members	Final review and viva preparation	Reviewed final system, prepared demo flow and viva presentation

Appendix C : Log sheet (Attached separately)